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# Neural Network Certification Informed Power System Transient Stability Preventive Control with Renewable Energy

Tong Su, *Graduate Student Member, IEEE*, Junbo Zhao, *Senior Member, IEEE*

**Abstract**—Existing neural network-based surrogate modeling methods for transient stability constrained-optimal power flow (TSC-OPF) lack certifications in the presence of unseen perturbations or uncertainties. This may lead to divergence of TSC-OPF or insecure control strategies. This paper proposes a neural network certification-informed power system transient stability preventive control method considering the impacts of various uncertainty resources, such as errors from measurements, fluctuations in renewable energy sources (RESs) and loads, etc. A deep belief network (DBN) is trained to estimate the transient stability, replacing the time-consuming time-domain simulation-based calculations. Then, DBN is embedded into the iterations of the primal-dual interior-point method to solve TSC-OPF. To guarantee the robustness of the solutions, the neural network verifier  $\alpha, \beta$ -CROWN to deal with uncertainties from RESs and loads is proposed. The yielded certification results allow us to further adjust the transient stability safety margin under the iterated TSC-OPF solution process, balancing system security and economics. Numerical results on a modified western South Carolina 500-bus system demonstrate that the proposed method can efficiently and quickly obtain the safety-verified preventive control strategy through RES curtailment and generator dispatch with only a slight increase in cost.

**Index Terms**—Neural network robustness verification, transient stability, preventive control, deep belief network, uncertainty,  $\alpha, \beta$ -CROWN.

## NOMENCLATURE

### Abbreviations

|           |                                         |
|-----------|-----------------------------------------|
| BaB       | Branch and bound                        |
| DAE       | Differential-algebraic equation         |
| DBN       | Deep belief network                     |
| (D/F/G)NN | (Deep/feedforward/graph) neural network |
| DoS       | Denial-of-service                       |
| FDIA      | False data injection attack             |
| IBR       | Inverter-based resource                 |
| LP        | Linear programming                      |
| LSTM      | Long short-term memory                  |
| MAE       | Mean absolute error                     |
| MCS       | Monte Carlo sampling                    |
| MIP       | Mixed-integer programming               |
| PDIPM     | Primal-dual interior-point method       |
| PGD       | Projected gradient descent              |

|         |                                                    |
|---------|----------------------------------------------------|
| RES     | Renewable energy source                            |
| RMS     | Restricted Boltzmann machine                       |
| SG      | Synchronous generator                              |
| TDS     | Time-domain simulation                             |
| TSC-OPF | Transient stability constrained-optimal power flow |
| TSI     | Transient stability index                          |

### Notations

|                        |                                                                                                                               |
|------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| $\alpha$               | Adjustable lower bound linear coefficients in $\alpha$ -CROWN                                                                 |
| $\beta$                | Lagrange multiplier                                                                                                           |
| $\delta/\omega$        | Rotor angle/speed                                                                                                             |
| $\delta^{\max}$        | Maximum rotor angle difference of any two generators during the TDS                                                           |
| $\epsilon$             | Perturbation range of $\mathcal{C}$                                                                                           |
| $\zeta$                | Convergence threshold in the bisection method                                                                                 |
| $\eta$                 | Step size                                                                                                                     |
| $\theta/v_m$           | Bus voltage angle/magnitude                                                                                                   |
| $\lambda$              | Boundary between transient stability and instability                                                                          |
| $\mu$                  | Control strategies: $\mu = [p_{IBR}, p_{SG}]^T$                                                                               |
| $\tilde{\mu}$          | $\mu$ with uncertainties: $\tilde{\mu} = [\tilde{p}_{IBR}, \tilde{p}_{SG}]^T$                                                 |
| $\Pi_{\mathcal{C}}$    | Projection operator to set $\mathcal{C}$                                                                                      |
| $\rho$                 | Physical parameters                                                                                                           |
| $\sigma$               | ReLU activation function                                                                                                      |
| $\tau$                 | Uncertainties caused by measurement and forecast errors, as well as fluctuations in RESs and loads                            |
| $\chi_t/\psi_t$        | State/algebraic variables                                                                                                     |
| $a$                    | Linear coefficient of the input $x$ in the bound propagation in $\beta$ -CROWN                                                |
| $a_{SG}/b_{SG}/c_{SG}$ | Fuel cost coefficients of $p_{SG}$                                                                                            |
| $a_{CROWN}/c_{CROWN}$  | Linear coefficients/constants of the linear relaxation lower bound in CROWN                                                   |
| $\mathcal{C}$          | $\epsilon$ -bounded $\ell_p$ norm ball centered at $\tilde{x}$ : $\mathcal{C} = \{x \mid \ x - \tilde{x}\ _p \leq \epsilon\}$ |
| $C_g$                  | Connection matrix of SGs and IBRs                                                                                             |
| $c$                    | Constant term in $\beta$ -CROWN                                                                                               |
| $D$                    | Diagonal matrix used for relaxing ReLU                                                                                        |
| $d_{IBR}$              | Curtailment cost coefficient of $p_{IBR}$                                                                                     |
| $F/G$                  | Differential/algebraic equation                                                                                               |
| $f$                    | NN function: $f(x) = \mathcal{NN}(x, W, b) = z^{(L)}$                                                                         |

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|                                                           |                                                                                                                                        |
|-----------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| $f^*$                                                     | Minimum of $f$ : $f^* = \min f(\mathbf{x}), \forall \mathbf{x} \in \mathcal{C}$                                                        |
| $\underline{f}$                                           | Relaxed lower bound of $f^*$ : $\underline{f} \leq f^*$                                                                                |
| $\bar{\mathcal{G}}_p/\mathcal{G}_q$                       | Active/reactive power balance equation                                                                                                 |
| $g(\beta)/g(\alpha, \beta)$                               | Optimization formulation in $\beta$ -CROWN                                                                                             |
| $\mathcal{L}(\mathbf{x}_{\text{adv}}, y)$                 | Loss function of NN/DBN                                                                                                                |
| $L$                                                       | Number of layers in the NN                                                                                                             |
| $\mathcal{M}_{\text{SG},t}/\mathcal{M}_{\text{IBR},t}$    | Cost of SG generation/IBR curtailment                                                                                                  |
| $\mathcal{P}_{\text{bus}}/\mathcal{Q}_{\text{bus}}$       | Bus active/reactive power injection equation                                                                                           |
| $\mathbf{P}$                                              | Matrix encoding neuron split constraints in $\beta$ -CROWN                                                                             |
| $p_{\text{IBR}}/p_{\text{SG}}$                            | Active power generation of IBRs/SGs                                                                                                    |
| $\tilde{p}_{\text{IBR}}/\tilde{p}_{\text{SG}}$            | $p_{\text{IBR}}/p_{\text{SG}}$ with uncertainties                                                                                      |
| $p_{\text{SG}}^{\text{max}}/p_{\text{SG}}^{\text{min}}$   | Upper/lower limit of $p_{\text{SG}}$                                                                                                   |
| $q_{\text{SG}}^{\text{max}}/q_{\text{SG}}^{\text{min}}$   | Upper/lower limit of $q_{\text{SG}}$                                                                                                   |
| $\bar{p}_{\text{IBR}}$                                    | Forecasted maximum value of $p_{\text{IBR}}$                                                                                           |
| $q_{\text{IBR}}^{\text{max}}/q_{\text{IBR}}^{\text{min}}$ | Upper/lower limit of $q_{\text{IBR}}$                                                                                                  |
| $p_g/q_g$                                                 | Bus active/reactive power injection by SGs and IBRs                                                                                    |
| $p_d/q_d$                                                 | Bus active/reactive load demand                                                                                                        |
| $\tilde{p}_d/\tilde{q}_d$                                 | $p_d/q_d$ with uncertainties                                                                                                           |
| $p_l/p_l^{\text{max}}$                                    | Line active power flow/maximum value of $p_l$                                                                                          |
| $\mathbf{q}$                                              | Coefficients for adjusting neuron split constraints in $\beta$ -CROWN                                                                  |
| $\mathbf{S}$                                              | Diagonal matrix that encodes the split state of the neurons                                                                            |
| $s_{\text{IBR}}^{\text{rated}}$                           | Rated apparent power of IBRs                                                                                                           |
| TSI                                                       | Estimated TSI                                                                                                                          |
| $t$                                                       | Time instant                                                                                                                           |
| $u/l$                                                     | Upper/lower bound of intermediate layer $z$                                                                                            |
| $v_m^{\text{max}}/v_m^{\text{min}}$                       | Upper/lower limit of $v_m$                                                                                                             |
| $\mathbf{W}/\mathbf{b}$                                   | Weights/biases of NN/DBN                                                                                                               |
| $\mathbf{x}$                                              | Input vector of NN/DBN: $\mathbf{x} = [p_{\text{IBR}}, p_{\text{SG}}, p_d, q_d]^\top$                                                  |
| $\tilde{\mathbf{x}}$                                      | $\mathbf{x}$ with uncertainties: $\tilde{\mathbf{x}} = [\tilde{p}_{\text{IBR}}, \tilde{p}_{\text{SG}}, \tilde{p}_d, \tilde{q}_d]^\top$ |
| $\mathbf{x}_{\text{adv}}$                                 | Adversarial examples                                                                                                                   |
| $y$                                                       | Correct label, i.e., TDS-based TSI                                                                                                     |
| $\mathcal{Z}$                                             | Set of the split constraints on $z$ : $\mathcal{Z} = \mathcal{Z}^{(1)} \cap \dots \cap \mathcal{Z}^{(L-1)}$                            |
| $\mathcal{Z}^{+(i)}/\mathcal{Z}^{-(i)}$                   | Set of neuron indices with positive/negative split constraints in layer $i$                                                            |
| $z^{(i)}/\hat{z}^{(i)}$                                   | Pre-activation/post-activation value in NN                                                                                             |

## I. INTRODUCTION

RECENTLY, NNs have achieved great success in various power system applications, such as RES forecasting [1], load forecasting [2], stability estimation [3], and control [4]. However, the reliability and robustness of NNs have raised concerns, as they are vulnerable to adversarial examples, meaning that imperceptible perturbations of test samples might unexpectedly change the NNs estimations [5]–[7].

For practical deployments of NNs in power systems, the inputs to NNs are typically measurements or forecasts. The measurement errors associated with different measurement devices can affect the estimations made by NNs [8], [9]. Moreover,

measurement data may be subjected to false data injections and cyber-attacks [10], [11]. The adversarial samples generated in this way can be very close to true values but may lead to opposite NN estimations, such as misjudging an unstable scenario as stable, failing to trigger necessary controls, leading to system failures, blackouts, or even more severe accidents [12], [13]. In addition, the forecasts of RESs or loads have high uncertainty, and control strategies based on deterministic forecasts may be unsafe when executed. Therefore, robust estimation and control considering all perturbations are crucial.

There has been some research on the robustness analysis of NNs, which can be categorized into adversarial sample generation, adversarial sample-based NNs' robustness enhancement, and verification of NNs' robustness. The core idea for generating adversarial samples is similar, which is to set neighborhoods with different norms based on the samples that need to be verified. The adversarial samples are situated throughout the entire neighborhood, and the size of this neighborhood is determined by the perturbation amplitude of the samples. Then, these adversarial samples are used to retrain the NNs, enhancing their robustness. For example, [14] proposes a state-perturbation-based adversarial example and the false data injection attacks method that can carry out attacks stealthily to both conventional bad data detectors and deep learning-based detectors. In [15], a signal-specific method and a universal signal-agnostic method are proposed, which can respectively generate perturbations to misclassify most natural signals with high probability and have a higher transfer rate of black-box attacks. In addition, adversarial training is adopted to defend systems against adversarial attacks in [15]. Robustness verification is divided into two types, one is based on MCS to find the largest neighborhood, where the sample being verified will not be misclassified. A larger neighborhood around a test sample indicates greater robustness. In [13], the MCS-based method is applied to calculate the Lipschitz constant within a norm ball to approximate the robust index of NNs. Similarly, [16] employs a randomized smoothing algorithm to transform any classifier, which performs well under Gaussian noise, into a new classifier that is certifiably robust against adversarial examples. Given that it is not feasible to precisely calculate the classification probabilities of the base classifier, and it is also challenging to accurately assess the prediction and robustness ability of the smoothed classifier, the MCS is employed for both tasks, ensuring success with arbitrarily high probability. Then, based on sampling or probabilistic methods, it's impossible to guarantee with 100% certainty that all points within the neighborhood are reliably classified, leaving a risk of misclassification. Another category is analytical methods, for example, [17] calculates the norm distance from samples to the separating affine hyperplane as an index of robustness through the linearization of the model. However, for non-linear and non-convex models, achieving the optimal solution is challenging regardless of linearization. There may exist smaller distances to the separating affine hyperplane that lead to misclassification. [18] uses MIP to compute the minimum distance to a sample which changes the classification, where the ReLU function is piecewise linearized through binary variables. While MIP has proven effective in verifying the

robustness of NNs, particularly those employing piecewise linear activation functions such as ReLU, they encounter considerable obstacles when applied to NNs featuring more complex and general nonlinearities. Moreover, MIP is highly time-consuming for verifying large networks. In addition, [19] analyzes the threat of adversarial attacks on NN models and proposes a causality-based mitigation strategy, which avoids the high computational cost of traditional adversarial training. However, it only modifies the optimization objective and training process of the NN, without fully guaranteeing security or verifying robustness.

To rigorously ensure that the robustness within the  $p$ -norm ball of the uncertain example is fully verified, and applicable to general activation functions, the CROWN is proposed [20]. CROWN is an efficient bound propagation-based verification algorithm that backpropagates a linear inequality through the network, relaxing activation functions with linear bounds. Additionally, by adaptively selecting the linear approximation when computing certified lower bounds of minimum adversarial distortion, it can improve the certified lower bound [20].  $\alpha$ -CROWN enhances the CROWN verifier by optimizing both intermediate and final layer bounds using variable  $\alpha$ , offering superior effectiveness over LP by more efficiently tightening intermediate layer bounds [21]. Then,  $\beta$ -CROWN extends the CROWN verifier by integrating ReLU split constraints in BaB into the bound propagation process to optimize parameter  $\beta$  [22]. [23] extends BaB-based verification for non-ReLU and general nonlinear functions, achieving significant improvements in verifying NNs with non-ReLU activation functions, such as Transformer and long short-term memory networks. Furthermore, GPUs can effectively parallelize and accelerate bound propagation in the BaB process for  $\beta$ -CROWN and GCP-CROWN [24].

Preventive control aims to prepare the system for potential credible contingencies through generator dispatch and RES curtailment [25]. TSC-OPF is widely used to find the optimal operating point under transient stability constraints, but it relies on time-consuming TDSs, making it challenging for online applications in large-scale power systems. Several alternatives to TDSs are proposed to accelerate the calculation, such as transient energy function method [26], extended equal-area criterion [27], trajectory sensitivity method [28], etc. Deep learning methods have been applied as surrogate models for TDSs to estimate power system transient stability, achieving significant acceleration from several to tens of seconds to less than 0.01 seconds [29]. For example, in [30], the DBN model was used as a surrogate for TDSs, significantly accelerating the transient stability estimation. It was then embedded into the NSGA-III algorithm to search for the optimal preventive control strategy under multiple objectives. However, due to measurement or forecast errors, as well as cyber-attacks, these deterministic NN-based control methods cannot guarantee the robustness of the strategy. To address this issue, [31] trains a Bayesian NN for probabilistic transient stability estimation, while Bayesian optimization was used to efficiently solve the security-constrained optimal power flow problem. Furthermore, [29] embeds a probabilistic Gaussian process model in TSC-OPF to analytically determine preven-

tive control strategies. However, these approaches rely solely on the assumption that the probabilistic NN output follows a normal distribution. This has two main limitations: first, the results contain inherent errors; second, they only account for uncertainties in the existing data, making it difficult to incorporate future uncertainties. Therefore, developing robust verification methods that ensure the reliability of all possible scenarios within the uncertainty range is critical. However, most current NN verification research focuses on improving the robustness of NN-based estimation results and is still largely based on MCSs, which do not provide complete verification. As a result, for high-dimensional, nonlinear, and non-convex transient stability control problems, there is currently no fast and complete NN verification method.

This paper proposes an NN robustness certification informed power system transient stability preventive control method. The main contributions are summarized as follows:

- 1) NNs are widely used in power system estimation and control, yet research on their robustness verification remains limited. Most existing studies rely on incomplete Monte Carlo-based methods, which do not verify every scenario within the estimated uncertainty interval. In contrast, a complete verification ensures that all possible scenarios are validated. In this paper, to guarantee the robustness of the control solutions, the complete NN verifier  $\alpha, \beta$ -CROWN is introduced for the first time in the power system transient stability estimation problem to address measurement or forecast errors as well as the uncertainty of RESs and loads.
- 2) In transient stability control, no studies have incorporated NN verification. For the first time,  $\alpha, \beta$ -CROWN-based complete NN robustness verification method has been proposed for transient stability preventive control problem. The robustness of the DBN estimation results is verified to iteratively guide the adjustment of the transient stability safety margin in TSC-OPF. The proposed method effectively balances system security and economic efficiency while being scalable to large-scale systems. Additionally, the scalability of  $\alpha, \beta$ -CROWN to other NN models and architectures, as well as its applications in power system estimation and control, is discussed.

The rest of the paper is organized as follows. Section II presents the TSC-OPF problem. Section III introduces the proposed NN robustness verification-based preventive control framework. Results are presented and analyzed in Section IV, and finally, Section V concludes the paper.

## II. PROBLEM STATEMENT

TSC-OPF is applied to determine the optimal operating point subject to static and transient stability constraints, and its mathematical formulation is as follows [30].

### A. Objective Function

The objective function is to minimize the SG generation cost  $\mathcal{M}_{SG,t}$  and RES curtailment cost  $\mathcal{M}_{IBR,t}$ :

$$\begin{aligned} \min \quad & \mathcal{M}_{SG,t}(\mathbf{p}_{SG,t}) + \mathcal{M}_{IBR,t}(\mathbf{p}_{IBR,t}) \\ \mathcal{M}_{SG,t}(\mathbf{p}_{SG,t}) = & \sum \mathbf{a}_{SG} \mathbf{p}_{SG,t}^2 + \mathbf{b}_{SG} \mathbf{p}_{SG,t} + \mathbf{c}_{SG} \\ \mathcal{M}_{IBR,t}(\mathbf{p}_{IBR,t}) = & \sum \mathbf{d}_{IBR} (\bar{\mathbf{p}}_{IBR,t} - \mathbf{p}_{IBR,t}) \end{aligned} \quad (1)$$

### B. Power Flow Equations

The power flow equations include active power balance equation  $\mathcal{G}_p$  and reactive power balance equation  $\mathcal{G}_q$ :

$$\mathcal{G}_p(\boldsymbol{\theta}, \mathbf{v}_m, \mathbf{p}_g) = \mathcal{P}_{bus}(\boldsymbol{\theta}, \mathbf{v}_m) + \mathbf{p}_d - \mathbf{C}_g \mathbf{p}_g = \mathbf{0} \quad (2)$$

$$\mathcal{G}_q(\boldsymbol{\theta}, \mathbf{v}_m, \mathbf{q}_g) = \mathcal{Q}_{bus}(\boldsymbol{\theta}, \mathbf{v}_m) + \mathbf{q}_d - \mathbf{C}_g \mathbf{q}_g = \mathbf{0} \quad (3)$$

### C. Static Constraints

Static constraints include line capacity constraints, bus voltage magnitude constraints, SG active and reactive power generation constraints, IBRs active and reactive power generation constraints.

$$|\mathbf{p}_{l,t}| \leq \mathbf{p}_l^{\max} \quad \mathbf{v}_m^{\min} \leq \mathbf{v}_{m,t} \leq \mathbf{v}_m^{\max} \quad (4)$$

$$\mathbf{p}_{SG}^{\min} \leq \mathbf{p}_{SG,t} \leq \mathbf{p}_{SG}^{\max} \quad \mathbf{q}_{SG}^{\min} \leq \mathbf{q}_{SG,t} \leq \mathbf{q}_{SG}^{\max} \quad (5)$$

$$0 \leq \mathbf{p}_{IBR,t} \leq \bar{\mathbf{p}}_{IBR,t} \leq \mathbf{s}_{IBR}^{\text{rated}} \quad \mathbf{q}_{IBR}^{\min} \leq \mathbf{q}_{IBR,t} \leq \mathbf{q}_{IBR}^{\max} \quad (6)$$

$$\mathbf{q}_{IBR}^{\max/\min} = +/ - \sqrt{(\mathbf{s}_{IBR}^{\text{rated}})^2 - (\mathbf{p}_{IBR})^2} \quad (7)$$

### D. Transient Stability Constraint

Transient stability refers to the ability of the power system to maintain synchronism after being subjected to a severe disturbance [32], where the TSI can represent rotor angle stability through solving a series of DAEs. Based on this, the transient stability constraint is:

$$\begin{cases} \dot{\chi}_t = \mathcal{F}(\chi_t, \psi_t, \boldsymbol{\mu}, \boldsymbol{\rho}, \boldsymbol{\tau}) \\ \mathbf{0} = \mathcal{G}(\chi_t, \psi_t, \boldsymbol{\mu}, \boldsymbol{\rho}, \boldsymbol{\tau}) \end{cases} \quad (8)$$

$$\text{TSI}(\chi) = \frac{360^\circ - \delta^{\max}}{360^\circ + \delta^{\max}} \times 100 > \lambda \geq 0 \quad (9)$$

where state variables  $\chi_t$  include, such as the rotor angle  $\delta$  and rotor speed  $\omega$  of SGs, virtual rotor speeds and angles of IBRs, etc; algebraic variables  $\psi_t$  include, such as voltage and current. TSI is positively correlated with the system's transient stability, and it requires TSI to be greater than 0, meaning  $\text{TSI} > \lambda \geq 0$  [33]. Due to uncertainties  $\boldsymbol{\tau}$ , the scenario during the execution of the preventive control strategy may deviate from the one used to generate the strategy, potentially leading to control failure. Therefore,  $\lambda$  generally needs to be greater than 0 to ensure a certain margin, and the robustness of the control strategies needs to be verified.

## III. PROPOSED NN ROBUSTNESS VERIFICATION-BASED POWER SYSTEM PREVENTIVE CONTROL

In this section, we first introduce an advanced NN-based approach, DBN, for fast transient stability estimation. However, due to various uncertainties, misclassification may occur, necessitating robustness verification. We then introduce the traditional PGD attack for adversarial sample search, followed by an explanation of the limitations of the conventional

LP verifier. Next, we present the NN verification methods CROWN,  $\alpha$ -CROWN, and  $\beta$ -CROWN. Finally, we propose a transient stability preventive control method based on  $\alpha$ ,  $\beta$ -CROWN, which certifies the robustness of the DBN-based transient stability estimation model and ensures that TSC-OPF converges to the stability boundary, achieving optimal economic performance.

### A. DBN-Based Transient Stability Estimation

A DBN is an advanced NN model consisting of multiple stacked RBMs followed by a fully connected layer. Unlike conventional NNs trained end-to-end with backpropagation, DBN employs a two-stage training process: first, an unsupervised pretraining phase, where each RBM layer is trained sequentially to capture hierarchical feature representations, and then a supervised fine-tuning phase, where the pretrained RBMs are stacked into a DNN, and the entire model is refined using backpropagation, as detailed in [30]. This layer-wise pretraining helps prevent vanishing gradients, improves weight initialization, and enhances estimation accuracy. After training, a  $L$ -layer DBN functions like a FNN and follows the standard forward propagation formulation:

$$\begin{aligned} f(\mathbf{x}) &= \mathcal{NN}(\mathbf{x}, \mathbf{W}, \mathbf{b}) = \mathbf{z}^{(L)} \\ \mathbf{z}^{(L)} &= \mathbf{W}^{(L)} \hat{\mathbf{z}}^{(L-1)} + \mathbf{b}^{(L)}, \quad \hat{\mathbf{z}}^{(L)} = \mathbf{z}^{(L)} \\ \mathbf{z}^{(i)} &= \mathbf{W}^{(i)} \hat{\mathbf{z}}^{(i-1)} + \mathbf{b}^{(i)}, \quad \hat{\mathbf{z}}^{(i)} = \sigma(\mathbf{z}^{(i)}), \\ \hat{\mathbf{z}}^{(0)} &= \mathbf{x}, \quad i \in \{1, \dots, L-1\} \end{aligned} \quad (10)$$

where the input vector  $\mathbf{x}$  of DBN is  $\mathbf{x} = [\mathbf{p}_{IBR}, \mathbf{p}_{SG}, \mathbf{p}_d, \mathbf{q}_d]^\top$ . The output variable  $\mathbf{z}^{(L)}$  is the estimated TSI, i.e., TSI. By generating a large number of samples to train the DBN, the complex DAEs in (8) and (9) can be replaced. The DBN-based transient stability constraint is

$$\overline{\text{TSI}} = \text{DBN}(\tilde{\mathbf{x}}, \mathbf{W}, \mathbf{b}) > \lambda \geq 0 \quad (11)$$

where  $\tilde{\mathbf{x}}$  considers uncertainties compared to  $\mathbf{x}$ , including control strategies  $\tilde{\mathbf{p}}_{IBR}$  and  $\tilde{\mathbf{p}}_{SG}$ , as well as the forecasted  $\tilde{\mathbf{p}}_d$  and  $\tilde{\mathbf{q}}_d$ . By using (11), the speed of transient stability assessment can be improved by hundreds or even thousands of times. However, due to the presence of uncertainty, the robustness of the DBN evaluation results needs to be verified.

### B. Projected Gradient Descent Attack

PGD is a widely used iterative adversarial attack, particularly effective in evaluating the robustness of NN models. The goal of a PGD attack is to generate adversarial examples  $\mathbf{x}_{\text{adv}}$  that are sufficiently close to the original input  $\tilde{\mathbf{x}}$  to attack the NN, aiming to find points in the input space that, with small perturbations, can lead the NN to output incorrect classification results. PGD achieves this by iteratively maximizing the network's loss function  $\mathcal{L}(\mathbf{x}_{\text{adv}}, y)$  (where  $y$  is the correct label, i.e., TDS-based TSI). At each iteration, the adversarial example  $\mathbf{x}_{\text{adv}}$  is updated as follows:

$$\mathbf{x}_{\text{adv}}^{t+1} = \Pi_{\mathcal{C}}(\mathbf{x}_{\text{adv}}^t + \eta \cdot \text{sign}(\nabla_{\mathbf{x}} \mathcal{L}(\mathbf{x}_{\text{adv}}^t, y))) \quad (12)$$

where set  $\mathcal{C}$  defines the possible perturbation region of  $\tilde{\mathbf{x}}$ , which is an  $\epsilon$ -bounded  $\ell_p$  norm ball centered at  $\tilde{\mathbf{x}}$ , i.e.,  $\mathcal{C} = \{\mathbf{x} \mid \|\mathbf{x} - \tilde{\mathbf{x}}\|_p \leq \epsilon\}$ . Projection operator  $\Pi_{\mathcal{C}}$  ensuring that the updated  $\mathbf{x}_{\text{adv}}^t$  stays within the set  $\mathcal{C}$ . If there exists an  $\mathbf{x}_{\text{adv}}$  within  $\mathcal{C}$  that leads to incorrect classification, it indicates



the presence of an adversarial sample within  $\mathcal{C}$ . This suggests that, at the current level of uncertainty, a security risk exists. PGD attacks are fast but cannot provide global robustness guarantees, only revealing the NN's vulnerability at certain points.

### C. Linear Programming Verifier

Adversarial attacks, such as PGD, expose the vulnerabilities of NNs by identifying perturbed inputs that cause misclassification. However, they do not guarantee robustness across the entire perturbation set  $\mathcal{C}$ . To address this limitation, NN verification methods have been developed to formally certify whether adversarial examples exist within  $\mathcal{C}$ . For a standard and general NN model in (10), the NN verification can be formulated as an optimization problem as follows [22]:

$$\begin{aligned} \min \quad & f(\mathbf{x}) := \mathcal{NN}(\mathbf{x}, \mathbf{W}, \mathbf{b}) = \mathbf{z}^{(L)} \\ \text{s.t.} \quad & \mathbf{z}^{(i)} = \mathbf{W}^{(i)} \hat{\mathbf{z}}^{(i-1)} + \mathbf{b}^{(i)}, \quad \hat{\mathbf{z}}^{(i)} = \sigma(\mathbf{z}^{(i)}), \quad (13) \\ & \mathbf{x} \in \mathcal{C}, \quad i \in \{1, \dots, L-1\} \end{aligned}$$

For the NN input  $\tilde{\mathbf{x}}$ , with  $\text{TSI} = f(\tilde{\mathbf{x}}) > 0$ , if it can be proven that  $f(\mathbf{x}) > 0, \forall \mathbf{x} \in \mathcal{C}$ , then  $\tilde{\mathbf{x}}$  is successfully verified. NN verification is categorized into complete and incomplete verification. Complete verifiers can solve (13) exactly, i.e.,  $f^* = \min f(\mathbf{x}), \forall \mathbf{x} \in \mathcal{C}$ . In contrast, incomplete verifiers typically relax the non-convexity of NNs to obtain a tractable lower bound of the solution, i.e.,  $\underline{f} \leq f^*$ . If  $\underline{f} \geq 0$ , then  $f^* \geq 0$  so  $\tilde{\mathbf{x}}$  is verified, as shown in Fig. 1-a. However, if  $\underline{f} < 0$ , the verification result is unknown, as shown in Fig. 1-b.

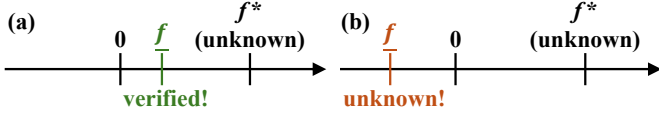


Fig. 1. Incomplete verification. (a):  $\underline{f} \geq 0$ ,  $\tilde{\mathbf{x}}$  is verified. (b):  $\underline{f} < 0$ , the verification result is unknown.

A commonly used incomplete verifier is the LP verifier, which relaxes non-convex ReLU constraints with linear constraints and transforms (13) into a LP problem. For  $\sigma(z_j^{(i)}) := \max(0, z_j^{(i)})$  and its intermediate layer bounds  $l_j^{(i)} \leq z_j^{(i)} \leq u_j^{(i)}$ , each ReLU can be relaxed by the “triangle” relaxation: (1) if  $l_j^{(i)} \geq 0$ ,  $\hat{z}_j^{(i)} = z_j^{(i)}$  (Fig. 2-a); (2) if  $l_j^{(i)} \leq 0$ ,  $\hat{z}_j^{(i)} = 0$  (Fig. 2-b); (3) if  $l_j^{(i)} \leq 0 \leq u_j^{(i)}$ , three linear bounds are used:  $\hat{z}_j^{(i)} \geq 0$ ,  $\hat{z}_j^{(i)} \geq z_j^{(i)}$ , and  $\hat{z}_j^{(i)} \leq \frac{u_j^{(i)}}{u_j^{(i)} - l_j^{(i)}} (z_j^{(i)} - l_j^{(i)})$  (Fig. 2-c). However, LP verifiers use fixed intermediate bounds and cannot use the joint optimization of intermediate layer bounds to tighten relaxation [22].

### D. CROWN and $\alpha$ -CROWN

Another cheaper way to find  $\underline{f}$  is CROWN, which backward propagates a linear bound from  $f$  to each intermediate layer, and finally to the input  $\mathbf{x}$ . CROWN uses a linear upper bound  $\hat{z}_j^{(i)} \leq \frac{u_j^{(i)}}{u_j^{(i)} - l_j^{(i)}} (z_j^{(i)} - l_j^{(i)})$  and a linear lower bound  $\hat{z}_j^{(i)} \geq$

$\alpha_j^{(i)} z_j^{(i)}$  ( $0 \leq \alpha_j^{(i)} \leq 1$ ) to relax non-convex ReLU, as shown in Fig. 2-d. The ReLU relaxation in CROWN is

$$\mathbf{W}^\top \sigma(\mathbf{z}) \geq \mathbf{W}^\top \mathbf{D} \mathbf{z} + \mathbf{b}' \quad (14)$$

where diagonal matrix  $\mathbf{D}$  is defined as:  $D_{j,j} = 1$  (if  $l_j \geq 0$ ),  $D_{j,j} = 0$  (if  $u_j \leq 0$ ),  $D_{j,j} = \alpha_j$  (if  $l_j < 0 < u_j$  and  $W_j \geq 0$ ), and  $D_{j,j} = \frac{u_j}{u_j - l_j}$  (if  $l_j < 0 < u_j$  and  $W_j < 0$ );  $\mathbf{b}' = \mathbf{W}^\top \mathbf{b}$  and  $\mathbf{b}_j = 0$  (if  $l_j > 0$  or  $u_j \leq 0$  or ( $l_j < 0 < u_j$  and  $W_j \geq 0$ )), and  $\mathbf{b}_j = -\frac{u_j l_j}{u_j - l_j}$  (if  $l_j < 0 < u_j$  and  $W_j < 0$ ). The CROWN can be formulated as [20]:

$$f_{\text{CROWN}}^* = \min_{\mathbf{x} \in \mathcal{C}} f(\mathbf{x}) \geq \min_{\mathbf{x} \in \mathcal{C}} \mathbf{a}_{\text{CROWN}}^\top \mathbf{x} + c_{\text{CROWN}} \quad (15)$$

where  $\mathbf{a}_{\text{CROWN}}$  and  $c_{\text{CROWN}}$  can be calculated using  $\mathbf{W}^{(i)}$ ,  $\mathbf{b}^{(i)}$ ,  $\mathbf{l}^{(i)}$  and  $\mathbf{u}^{(i)}$  in polynomial time. (15) can be solved using Hölder's inequality and accelerated by GPUs and TPUs [24]. Based on CROWN,  $\alpha$ -CROWN further uses gradient ascent to optimize  $\alpha$  and tighten the lower bound, as shown in Fig. 2-e. The  $\alpha$ -CROWN can be formulated as [21]:

$$f_{\alpha\text{-CROWN}}^* = \min_{\mathbf{x} \in \mathcal{C}} f(\mathbf{x}) \geq \max_{0 \leq \alpha \leq 1} \min_{\mathbf{x} \in \mathcal{C}} f_{\text{CROWN}}(\mathbf{x}; \alpha) \quad (16)$$

However, CROWN and  $\alpha$ -CROWN are incomplete verification methods because they involve a series of relaxations, which may also lead to unknown cases, as shown in Fig. 1-b.

### E. $\beta$ -CROWN

BaB method is widely used in complete verifiers by dividing the  $\mathcal{C}$  into two linear subdomains  $\mathcal{C}_1 = \{x \in \mathcal{C}, z_j^{(i)} \geq 0\}$  and  $\mathcal{C}_2 = \{x \in \mathcal{C}, z_j^{(i)} < 0\}$ . Then incomplete verifiers can be used to relax and estimate the lower bound of each subdomain. Since ReLU is linear, no relaxation is needed. If the lower bound for  $\mathcal{C}_i$  is greater than 0, i.e.,  $\underline{f}_{\mathcal{C}_i} > 0$ ,  $\mathcal{C}_i$  is verified; otherwise,  $\mathcal{C}_i$  needs to further branch other unstable ReLUs until it is verified, as shown in Fig. 3 [22].

The neuron splitting-based NN verification can be formulated as an optimization problem:

$$\min_{\mathbf{x} \in \mathcal{C}, \mathbf{z} \in \mathcal{Z}} f(\mathbf{x}) = \min_{\mathbf{x} \in \mathcal{C}, \mathbf{z} \in \mathcal{Z}} \mathbf{W}^{(L)} \hat{\mathbf{z}}^{(L-1)} + \mathbf{b}^{(L)} \quad (17)$$

Since  $\hat{\mathbf{z}}^{(L-1)} = \sigma(\mathbf{z}^{(L-1)})$ , use (14) to relax ReLU neuron at layer  $L-1$  to obtain:

$$\min_{\mathbf{x} \in \mathcal{C}, \mathbf{z} \in \mathcal{Z}} f(\mathbf{x}) \geq \min_{\mathbf{x} \in \mathcal{C}, \mathbf{z} \in \mathcal{Z}} \mathbf{W}^{(L)} \mathbf{D}^{(L-1)} \mathbf{z}^{(L-1)} + c \quad (18)$$

Then, the Lagrange function is used to eliminate the constraint  $\mathbf{z}^{(L-1)} \in \mathcal{Z}^{(L-1)}$  by adding  $\boldsymbol{\beta}^{(L-1)\top} \mathbf{S}^{(L-1)} \mathbf{z}^{(L-1)}$ , where  $\boldsymbol{\beta}$  denotes the Lagrange multiplier;  $\mathbf{S}$  is a diagonal matrix that encodes the split state of the neurons:  $\mathbf{S}_{j,j}^{(i)} = -1$  (if split  $z_j^{(i)} \geq 0$ ),  $\mathbf{S}_{j,j}^{(i)} = +1$  (if split  $z_j^{(i)} < 0$ ), and  $\mathbf{S}_{j,j}^{(i)} = 0$  (if no split for  $z_j^{(i)}$ ):

$$\begin{aligned} \min_{\mathbf{x} \in \mathcal{C}, \mathbf{z} \in \mathcal{Z}} f(\mathbf{x}) &\geq \min_{\mathbf{x} \in \mathcal{C}} \max_{\substack{\mathbf{z}^{(L-2)} \in \tilde{\mathcal{Z}}^{(L-2)} \\ \boldsymbol{\beta}^{(L-1)} \geq \mathbf{0}}} \mathbf{W}^{(L)} \mathbf{D}^{(L-1)} \mathbf{z}^{(L-1)} \\ &\quad + \boldsymbol{\beta}^{(L-1)\top} \mathbf{S}^{(L-1)} \mathbf{z}^{(L-1)} + c \\ &\geq \max_{\boldsymbol{\beta}^{(L-1)} \geq \mathbf{0}} \min_{\substack{\mathbf{x} \in \mathcal{C} \\ \mathbf{z}^{(L-2)} \in \tilde{\mathcal{Z}}^{(L-2)}}} \left( \mathbf{W}^{(L)} \mathbf{D}^{(L-1)} \right. \\ &\quad \left. + \boldsymbol{\beta}^{(L-1)\top} \mathbf{S}^{(L-1)} \right) \mathbf{z}^{(L-1)} + c \end{aligned} \quad (19)$$

where the second inequality is based on the weak duality;  $\mathcal{Z} = \mathcal{Z}^{(1)} \cap \dots \cap \mathcal{Z}^{(L-1)}$ ;  $\mathcal{Z}^{(i)} := \{z^{(i)} \mid z_{j_1}^{(i)} \geq 0, z_{j_2}^{(i)} < 0, \forall j_1 \in \mathcal{Z}^{+(i)}, \forall j_2 \in \mathcal{Z}^{-(i)}\}$ ;  $\tilde{\mathcal{Z}}^{(i)} := \mathcal{Z}^{(1)} \cap \dots \cap \mathcal{Z}^{(i)}$ ;

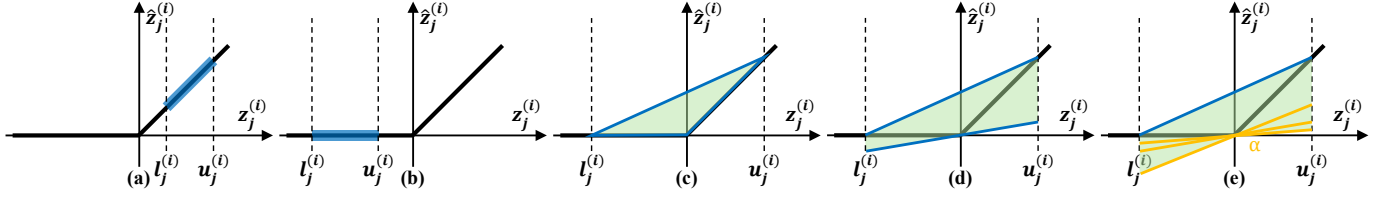


Fig. 2. Relaxations of a ReLU. (a)(b) No relaxation when  $l_j^{(i)} \geq 0$  or  $l_j^{(i)} \leq 0$ ; (c) “triangle” relaxation when  $l_j^{(i)} \leq 0 \leq u_j^{(i)}$  in LP verifiers; (d) linear relaxation in CROWN with fixed lower bounds; (e) linear relaxation in  $\alpha$ -CROWN with adjustable lower bounds.

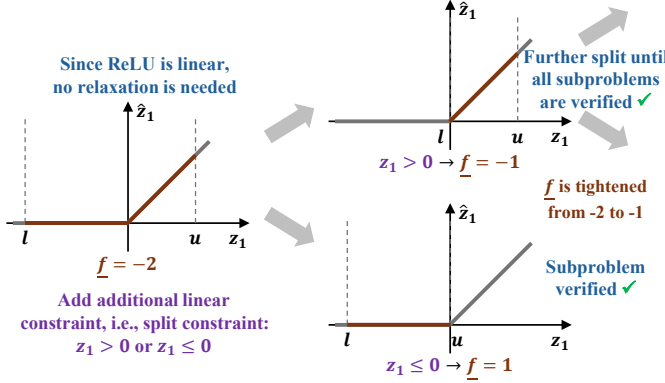


Fig. 3. BaB-based NN splitting and verification.

$\tilde{z}^{(i)} := \{z^{(1)}, z^{(2)}, \dots, z^{(i)}\}$ . Then  $z^{(L-1)}$  is substituted by  $\mathbf{W}^{(L-1)} \tilde{z}^{(L-2)}$  for next layer:

$$\min_{x \in \mathcal{C}, z \in \mathcal{Z}} f(x) \geq \max_{\beta^{(L-1)} \geq 0} \min_{\substack{x \in \mathcal{C} \\ \tilde{z}^{(L-2)} \in \tilde{\mathcal{Z}}^{(L-2)}}} \left( \mathbf{W}^{(L)} \mathbf{D}^{(L-1)} + \beta^{(L-1)\top} \mathbf{S}^{(L-1)} \right) \mathbf{W}^{(L-1)} \tilde{z}^{(L-2)} + c \quad (20)$$

Finally, by recursively back-substituting the linear constraints of the NN layer by layer and swapping max and min at each step, (20) becomes:

$$\min_{x \in \mathcal{C}, z \in \mathcal{Z}} f(x) \geq \max_{\beta \geq 0} \min_{x \in \mathcal{C}} (a + \mathbf{P}\beta)^\top x + \mathbf{q}^\top \beta + c \quad (21)$$

where  $\beta := [\beta^{(1)\top}, \beta^{(2)\top}, \dots, \beta^{(L-1)\top}]^\top$ ;  $a$ ,  $\mathbf{P}$ ,  $\mathbf{q}$ , and  $c$  are functions of  $\mathbf{W}^{(i)}$ ,  $\mathbf{b}^{(i)}$ ,  $\mathbf{l}^{(i)}$  and  $\mathbf{u}^{(i)}$ . In the  $\ell_p$  norm ball  $\mathcal{C} = \{x \mid \|x - \tilde{x}\|_p \leq \epsilon\}$ , the inner minimization has a closed solution:

$$\min_{x \in \mathcal{C}, z \in \mathcal{Z}} f(x) \geq \max_{\beta \geq 0} -\|a + \mathbf{P}\beta\|_q \epsilon + (\mathbf{p}^\top \tilde{x} + \mathbf{q}^\top \beta) + \mathbf{a}^\top \tilde{x} + c := \max_{\beta \geq 0} g(\beta) \quad (22)$$

where  $\frac{1}{p} + \frac{1}{q} = 1$ . The maximization is concave in  $\beta$  (when  $q \geq 1$ ) and can be optimized using projected (super) gradient ascent. Then, the slope  $\alpha = \{\alpha^{(1)}, \dots, \alpha^{(L-1)}\}$  of the ReLU lower bound is optimized to further tighten the bound, and (22) can be rewritten as:

$$\min_{x \in \mathcal{C}, z \in \mathcal{Z}} f(x) \geq \max_{0 < \alpha \leq 1, \beta \geq 0} g(\alpha, \beta) \quad (23)$$

#### F. $\alpha, \beta$ -CROWN-Based Transient Stability Preventive Control

Based on the aforementioned DBN-based transient stability estimation and a series of NN verification methods, this paper proposes for the first time a transient stability preventive control method with robustness verification using  $\alpha, \beta$ -CROWN.

#### Workflow 1: DBN Training for Transient Stability Classification (DBN-C) and Estimation (DBN-E).

**Input:** Historical dataset of  $x$

- 1 Set fault contingency.
- 2 Run TDS to calculate TSI.
- 3 Data preprocessing and normalization.
- 4 Build DBN-C and DBN-E in PyTorch, where DBN-C outputs the 0/1 classification labels for TSI, and DBN-E outputs the actual TSI.
- 5 Set the number of layers, the number of neurons per layer, the activation function ReLU, and other hyperparameters.
- 6 Set the cross entropy loss for DBN-C and the mean squared error loss for DBN-E, respectively.
- 7 Train DBN-C and DBN-E.

**Output:** DBN-C and DBN-E

This method leverages DBN to accelerate the transient stability assessment of preventive control strategies while verifying their robustness within the uncertainty set, ensuring both security and economic efficiency. The proposed  $\alpha, \beta$ -CROWN-based NN certification informed transient stability preventive control is divided into three parts:

#### 1) DBN Training for Transient Stability Classification and Estimation:

Two different DBN-based transient stability estimation models are trained, one for classification (DBN-C) and one for estimation (DBN-E). This is because  $\alpha, \beta$ -CROWN requires a classification model to verify whether the lower bound misclassifies into another category. Therefore, the output of DBN-C adopts a one-hot encoding format, where TSI is split into two dimensions: one representing transient stability and the other representing transient instability. When the system belongs to a specific category, its corresponding output is set to 1, while the other remains 0. In addition, the one-hot encoding format avoids the issue of threshold determination associated with single-output methods. However, TSC-OPF requires a DBN-based transient stability estimation surrogate model to replace TDS and estimate the actual TSI value. This is because TSC-OPF is solved using the PDIPM [34], which necessitates computing the Jacobian and Hessian matrices of each constraint to guide the iteration direction. For example, in the transient stability constraint (11), the Jacobian and Hessian matrices of TSI with respect to the control strategy  $\mu$  must be computed to determine which variables should be adjusted to increase TSI, ensuring a gradual transition from transient instability to stability during optimization. This cannot be

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**Algorithm 1:**  $\alpha, \beta$ -CROWN-Based Transient Stability Robustness Verification.

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**Input:**  $\tilde{x}$ , DBN-C

**Output:** Verification results: *Safety* = *safe* or *unsafe* or *unknown*

- 1 Set  $\mathcal{C}$  according to the possible perturbation range of  $\mathbf{p}_{\text{IBR}}$ ,  $\mathbf{p}_{\text{SG}}$ ,  $\mathbf{p}_d$ , and  $\mathbf{q}_d$ .
  - 2 Set *Safety* = *unknown*.
  - 3 Perform PGF attack according to (12).
  - 4 Compare the classification results before and after the PGD attack. If the classification changes, the attack is considered successful.
  - 5 **if** *PGD attack* == *True* **then**
  - 6     **Return:** *Safety* = *unsafe*
  - 7 **end**
  - 8 Perform  $\alpha$ -CROWN according to (16).
  - 9 **if** *Safety* == *unknown* **and**  $\alpha$ -CROWN == *True* **then**
  - 10     **Return:** *Safety* = *safe*
  - 11 **end**
  - 12 Perform  $\beta$ -CROWN according to (23).
  - 13 **if** *Safety* == *unknown* **and**  $\beta$ -CROWN == *True* **then**
  - 14     **Return:** *Safety* = *safe*
  - 15 **end**
  - 16 **Return:** *Safety* = *unknown*
- 

achieved using the classification model DBN-C, as it only provides discrete classification results rather than continuous values. Therefore, two DBN models are trained. The detailed steps of DBN training are shown in Workflow 1.

**2)  $\alpha, \beta$ -CROWN-Based Transient Stability Robustness**

**Verification:** Verify the preventive control strategy  $\mu$  using  $\alpha, \beta$ -CROWN for DBN-C. Due to the increasing verification time with PGD attacks,  $\alpha$ -CROWN, and  $\beta$ -CROWN, execute them sequentially. If a verification result is obtained, return immediately; otherwise, proceed to the next verification method. The results of NN verification are categorized into three types: safe (no misclassification exists within  $\mathcal{C}$ ), unsafe (there are misclassified points within  $\mathcal{C}$ ), and unknown (the robustness of  $\mathcal{C}$  cannot be verified). The verification steps are shown in Algorithm 1.

**3)  $\alpha, \beta$ -CROWN Certification Informed Transient Sta-**

**bility Preventive Control:** Execute TSC-OPF, where the DBN is embedded as a surrogate model for TDS to accelerate the transient stability estimation, as described in (11). This nonlinear and non-convex problem is solved using PDIPM [34], with the objective function given in (1), static constraints in (2) - (7), and transient stability constraint in (11). When solving the optimization problem, PDIPM requires computing the Jacobian and Hessian matrices of the objective function and constraint functions with respect to the optimization variables, i.e., the preventive control strategy  $\mu$ , to guide the update direction and step size. For the objective function and static constraints, computing their Jacobian and Hessian matrices with respect to  $\tilde{\mu}$  is straightforward, as explicit equations can be derived. For the DBN-E-based transient stability estimation model, the Jacobian and Hessian matrices of estimated

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**Algorithm 2:**  $\alpha, \beta$ -CROWN Certification Informed Transient Stability Preventive Control.

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**Input:**  $\tilde{\mathbf{p}}_{\text{IBR}}$ ,  $\tilde{\mathbf{p}}_d$ ,  $\tilde{\mathbf{q}}_d$ , DBN-E

**Output:** Safety-verified preventive control strategies  $\mu$ 

- 1 Initialize the  $\tilde{\mathbf{p}}_{\text{IBR}}$  and  $\tilde{\mathbf{p}}_{\text{SG}}$  as the operating scenario for the next preventive control time step.
  - 2 Set  $\lambda = 0$ ,  $\lambda_{\text{left}} = 0$ ,  $\lambda_{\text{right}} = 90$ ,  $\Delta\lambda = 2$ , and  $\zeta = 1$ .
  - 3 **while**  $|\Delta\lambda| \geq \zeta$  **do**
  - 4     Perform TSC-OPF using PDIPM according to the objective function (1), the static constraints (2) - (7), and the transient stability constraint (11).
  - 5     Obtain the preventive control strategy  $\tilde{\mu}$ .
  - 6     Verify  $\tilde{x} = [\tilde{\mu}, \tilde{\mathbf{p}}_d, \tilde{\mathbf{q}}_d]^\top$  according to Algorithm 1.
  - 7     **if** *Converge* == *True* **and** *Safety* != *safe* **then**
  - 8          $\Delta\lambda = \lambda_{\text{right}} - \lambda$ ,  $\lambda_{\text{left}} = \lambda$ ,  $\lambda = \lambda_{\text{left}} + \Delta\lambda/2$ .
  - 9     **end**
  - 10    **else if** *Converge* == *False* **and** *Safety* != *safe* **then**
  - 11       No feasible solution. Reduce the range of  $\mathcal{C}$  or implement other control measures, such as load shedding.
  - 12    **end**
  - 13    **else**
  - 14        $\Delta\lambda = \lambda - \lambda_{\text{left}}$ ,  $\lambda_{\text{right}} = \lambda$ ,  $\lambda = \lambda_{\text{left}} + \Delta\lambda/2$ .
  - 15    **end**
  - 16 **end**
- 

TSI with respect to  $\mathbf{x}$ , i.e.,  $d\text{TSI}/d\tilde{\mathbf{x}}$  and  $d^2\text{TSI}/d\tilde{\mathbf{x}}d\tilde{\mathbf{x}}^\top$ , can either be explicitly formulated based on the in (10) or computed using PyTorch's automatic differentiation function. Since  $\tilde{\mu}$  is a subset of  $\tilde{\mathbf{x}}$ , the required  $d\text{TSI}/d\tilde{\mu}$  and  $d^2\text{TSI}/d\tilde{\mu}d\tilde{\mu}^\top$  for PDIPM can be directly obtained from  $d\text{TSI}/d\tilde{\mathbf{x}}$  and  $d^2\text{TSI}/d\tilde{\mathbf{x}}d\tilde{\mathbf{x}}^\top$ . These Jacobian and Hessian matrices then guide the PDIPM iterations to obtain the optimal control strategy  $\mu$ . Afterward, the robustness of  $\mu$  is verified using the  $\alpha, \beta$ -CROWN described in the Algorithm 1. Since  $\lambda$  determines the safety margin of the transient stability constraint, different values of  $\lambda$  can be set to control the safety margin of  $\mu$ . The value of  $\lambda$  can be computed using the bisection method: if the verification result is safe, reduce  $\lambda$  (with a minimum of 0); otherwise, increase  $\lambda$  until the verification result is safe, resulting in the final safety-verified  $\mu$ . When TSC-OPF does not converge and the NN verification fails, the problem is unsolvable. The details are shown in Algorithm 2. The whole framework of the proposed method is shown in Fig. 4.

It is important to note that since this paper focuses on preventive control, only high-probability uncertainty distributions are considered for prevention. Additionally, as long as the disturbances fall within the defined uncertainty region, the method is applicable regardless of the underlying distribution — it is not limited to Gaussian distributions. In addition, cyber-attacks are categorized into various types, including FDIA, adversarial attacks, DoS attacks, and control signal tampering. Among these, adversarial attacks introduce small perturbations to mislead NNs, causing incorrect system stability estimations, which the proposed method can partially defend against. FDIA



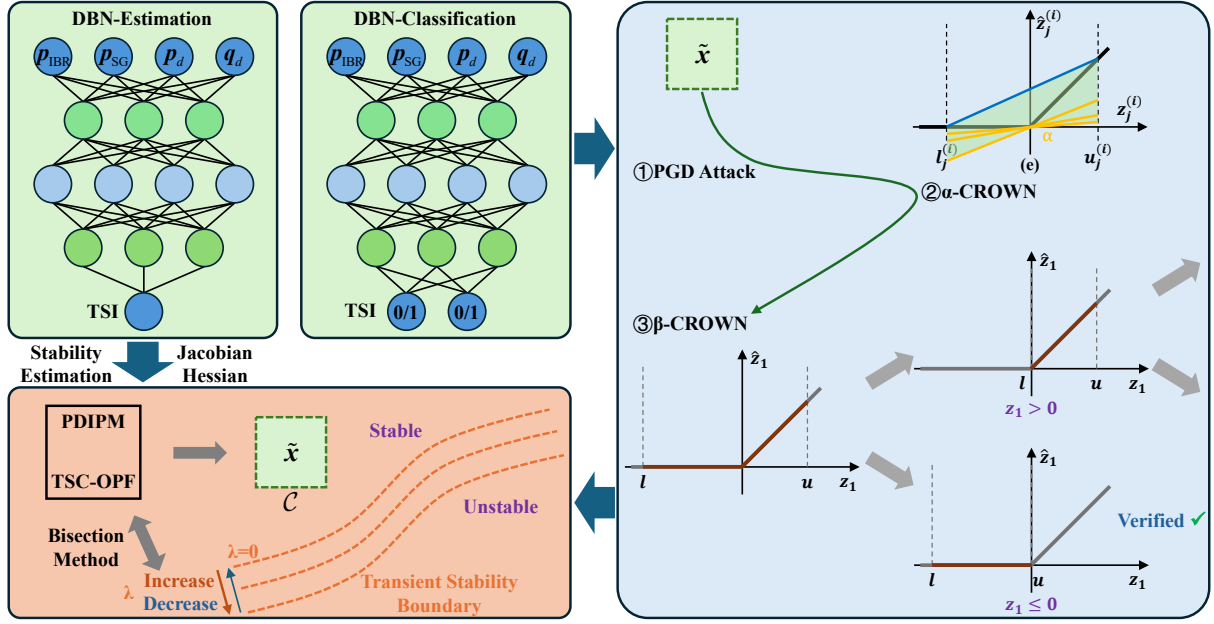


Fig. 4. Proposed  $\alpha, \beta$ -CROWN-based transient stability preventive control framework.

manipulates sensor or state estimation data, meaning that the proposed method can only defend against attacks that remain within  $\mathcal{C}$ . However, DoS and control signal tampering attacks fall completely outside the scope of this method and require additional defense mechanisms. Therefore, for large-scale FDIA, DoS attacks, and control signal tampering, it is recommended to integrate a dedicated attack detection process before and after applying the proposed method.

This subsection presents the use of  $\alpha, \beta$ -CROWN for verifying transient stability estimation results and its application in preventive control to improve economic efficiency. However, it is important to note that the proposed method only demonstrates one application of  $\alpha, \beta$ -CROWN-based NN verification in power systems. In fact, the  $\alpha, \beta$ -CROWN method is compatible with various NN layers, activation functions and models, including [22]:

- **Layers:** Fully connected, convolutional, pooling (average pooling and max pooling), and transposed convolution.
- **Activation functions and nonlinear operations:** ReLU, sigmoid, tanh, arctan, sin, cos, tan, GELU, power functions, multiplication, and self-attention.
- **Network structures:** Residual connections, Transformers, LSTMs, and GNNs.

In addition,  $\alpha, \beta$ -CROWN supports various types of  $p$ -norm balls, including [22]:

- $\ell_1$ : Suitable for scenarios where variables can substitute or compensate for each other.
- $\ell_2$ : Suitable for scenarios where variables are interrelated and require smooth overall optimization.
- $\ell_\infty$ : Suitable for scenarios where variables are independent of each other and the focus is on the worst-case scenario.

Therefore,  $\alpha, \beta$ -CROWN is applicable for robustness verification in various power system applications, including:

- Voltage and frequency stability assessment.
- LSTM or Transformer-based RES and load forecasting.
- Electric vehicle and battery charging and discharging optimization.
- GNN-based evaluation considering topology changes.
- Power system state estimation.
- Electricity market bidding and optimization.
- Cyber attack analysis.

In particular, the application of NN robustness verification in various power system optimization and control fields is almost nonexistent. Therefore, exploring more reasonable and efficient ways to integrate  $\alpha, \beta$ -CROWN into a broader range of applications is a promising direction.

#### IV. NUMERICAL RESULTS

Numerical results are tested on the modified South Carolina 500-bus system, which serves 21 counties and approximately 2.6 million people, with 206 loads supplied by 9 IBRs and 51 SGs, as shown in Fig. 5 [35].

##### A. $\alpha, \beta$ -CROWN-Based Transient Stability Robustness Verification

According to Workflow 1, two 5-layer DBN models are built in PyTorch for transient stability classification (DBN-C) and estimation (DBN-E), respectively. The classification accuracy of DBN-C is 99.15%, and the MAE of estimation by DBN-E is 1.86, demonstrating excellent accuracy. The detailed training and comparison results of DBN are shown in [30]. Additionally, we compared DBN-E with LSTM and Transformer, which are commonly used for time-series prediction. The results, shown in Table I, indicate that DBN-E outperforms the other models in terms of evaluation accuracy, MAE, and training time. This is because the problem focuses on single-point stability assessment and directly predicts TSI, rather



Fig. 5. One-line diagram of the South Carolina 500-bus system.

than trajectory prediction, where the advantages of LSTM and Transformer cannot be fully utilized. The classification or estimation time of DBN is less than 0.01s, and meets the requirement of online application and subsequent robustness verification and preventive control.

TABLE I  
PERFORMANCE COMPARISON BETWEEN DBN AND OTHER NNs

| Model       | Accuracy | MAE  | Training Time |
|-------------|----------|------|---------------|
| DBN         | 99.15%   | 1.86 | 148s          |
| LSTM        | 96.65%   | 9.28 | 167s          |
| Transformer | 98.95%   | 9.96 | > 1h          |

To demonstrate the transient stability robustness verification performance of  $\alpha, \beta$ -CROWN, 1,000 random samples are assumed as  $\tilde{x}$ , and their safety is verified using  $\alpha, \beta$ -CROWN. For power systems with element-wise bounds, the perturbation range  $\mathcal{C}$  is considered as an  $\ell_\infty$  norm ball around the forecasted  $\tilde{x}$ :  $\mathcal{C} = \{x \mid \|x - \tilde{x}\|_\infty \leq \epsilon\}$ . The maximum perturbation range for IBRs, SGs, and loads is set to 5% to construct  $\mathcal{C}$ . Then,  $\alpha, \beta$ -CROWN is used to test the probability that all points in  $\mathcal{C}$  maintain the same classification as  $\tilde{x}$ . The verification process is shown in Algorithm 1, and the verification results are shown in Table II. In Table II, unsafe-

TABLE II  
STATISTICAL PROBABILITIES OF NN ROBUSTNESS VERIFICATION RESULTS FOR RANDOM SCENARIOS

| unsafe-PGD | safe-incomplete | safe-complete | unknown |
|------------|-----------------|---------------|---------|
| 63.4%      | 29.7%           | 6.5%          | 0.4%    |

PGD indicates that the classification result changed due to the PGD attack; safe-incomplete means the verification result is classified as safe by  $\alpha$ -CROWN, while safe-complete means it is classified as safe by  $\beta$ -CROWN; unknown indicates that the verification result is still unknown after  $\alpha, \beta$ -CROWN. Since  $\alpha, \beta$ -CROWN progressively verifies robustness through PGD

attacks,  $\alpha$ -CROWN, and  $\beta$ -CROWN, 63.4% of the samples are verified as unsafe by PGD attacks, followed by 29.7% of the samples verified as safe by  $\alpha$ -CROWN. Finally, 6.9% of the samples needed verification by  $\beta$ -CROWN, among which 6.5% are found to be safe, and 0.4% are unsafe. From the verification results, it can be observed that  $\alpha, \beta$ -CROWN can obtain the robustness verification result for the preventive control strategy  $\mu$  in 99.6% of cases, and this result is complete. In addition, compared to methods that rely solely on PGD attacks, the verification success rate improved from 63.4% to 99.6% [19], [36].

To further verify the effects of different perturbations, the previously used 1,000 samples are tested with perturbations varying from 0% to 10%. The metrics tested include unsafe-PGD, safe-incomplete, safe-complete, and unknown, as shown in Fig. 6. From 6, it can be observed that when the perturbation

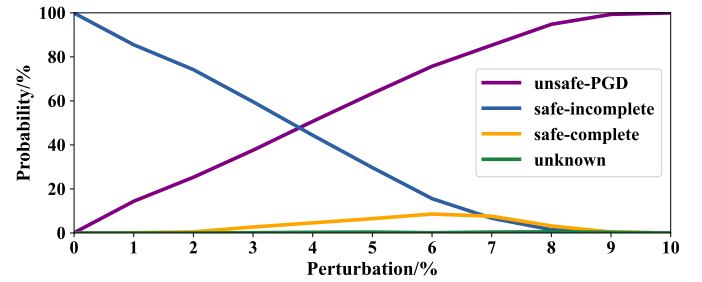


Fig. 6. Comparison of NN verification results with different perturbations.

is very small, the margin between  $\mathcal{C}$  and the transient stability boundary is large. In this case, PGD attacks are almost unable to determine whether the  $\tilde{x}$  is unsafe, while  $\alpha$ -CROWN can almost completely verify the robustness. As the perturbation increases, the margin between  $\mathcal{C}$  and the transient stability boundary decreases, allowing more  $\tilde{x}$  to be verified by PGD attacks. However, due to the looser lower bounds, incomplete verification via  $\alpha$ -CROWN may be unable to determine the robustness of some samples, necessitating complete NN verification through  $\beta$ -CROWN. As the perturbation grows closer to 10%,  $\mathcal{C}$  for most samples  $\tilde{x}$  crosses the transient stability boundary, allowing PGD attacks to verify the robustness. In addition, the probability of samples being verified as unknown is very low, consistently below 0.5%.

$\alpha, \beta$ -CROWN can also be used to determine the maximum possible perturbation and safety margin. For example, for a specific  $\tilde{x}$ , start with zero perturbation and gradually increase it up to 10%, verifying robustness at each step, as shown in Fig. 7. It is important to point out that safe-incomplete, safe-complete, and unsafe-PGD correspond to different meanings of the safety margin. Safe-incomplete refers to the distance from the  $\alpha$ -CROWN lower bound to 0, while unsafe-PGD corresponds to the difference between the values of two classifications. Safe-complete uses  $\beta$ -CROWN based on BaB, which further splits unstable ReLUs when the result is inconclusive, until the lower bound is greater than 0. Therefore, its final margin is a small positive value, but in practice, if all ReLUs are split, the margin could be further increased. Therefore, in Fig. 7, within the 0–5% perturbation range, as the perturbation increases, the safety margin between the  $\alpha$ -

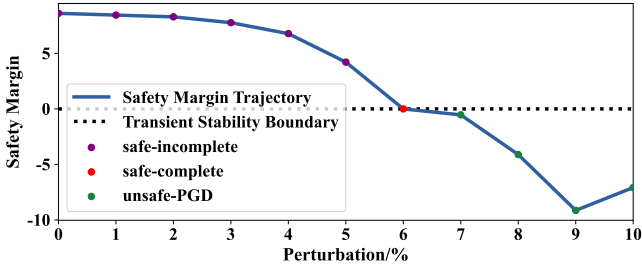


Fig. 7. Calculation of maximum possible perturbation and safety margin.

CROWN lower bound and the transient stability boundary gradually decreases, from 8.606 to 4.216. At 6% perturbation, the safety margin is  $1 \times 10^{-7}$ , meaning that splitting stopped once it became greater than 0. In the 7–10% perturbation range, the safety margin is always less than 0, indicating that the value corresponding to the classification of  $\tilde{x}$  is less than that of the other class. Since a 6% perturbation is classified as safe-complete and a 7% perturbation is classified as unsafe-PGD, the maximum possible perturbation for this particular  $\tilde{x}$  lies between 6% and 7%. If a more precise value is desired, further refinement between 6% and 7% can be performed. Additionally, the bisection method could be used to converge more quickly to this value. Since different verifications are completely decoupled, multiple servers can be used in parallel to determine this value more efficiently.

The average verification time for PGD attacks is only 0.00795s, while the average verification time for  $\alpha$ -CROWN and  $\beta$ -CROWN is 0.495s. The overall average verification time for  $\alpha, \beta$ -CROWN is 0.196s. From the verification times, it is evident that executing the PGD attack first can significantly save time. However, the verification times of all methods are very short, fully meeting the requirements of preventive control, which typically has a time interval of 15min or 1h. Additionally, due to its extremely fast evaluation time,  $\alpha, \beta$ -CROWN can also be extended to other real-time or near-real-time applications.

### B. $\alpha, \beta$ -CROWN-Based Transient Stability Preventive Control

A TSC-OPF without NN verification is performed to find the optimal  $\lambda$ . The preventive control strategy  $\mu$  for the active power generation of IBRs and SGs, compared to the case without preventive control, is shown in Fig. 8. In Fig. 8, the

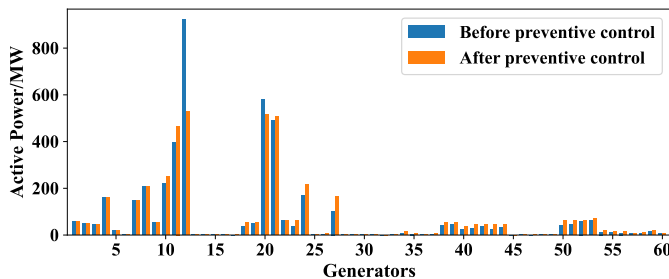


Fig. 8. Comparison of active power generation of IBRs and SGs before and after preventive control without NN robustness verification.

first 9 generators are IBRs, and the remaining 51 generators are SGs. It can be observed that all IBRs maintain the same active power output as  $\bar{p}_{IBR}$ , indicating no RES curtailment, thus minimizing cost. Meanwhile, the active power generations of all SGs are dispatched to minimize fuel costs. The total cost is \$19,035.6. The rotor angle trajectories before and after preventive control are shown in Fig. 9. From Fig. 9, it can

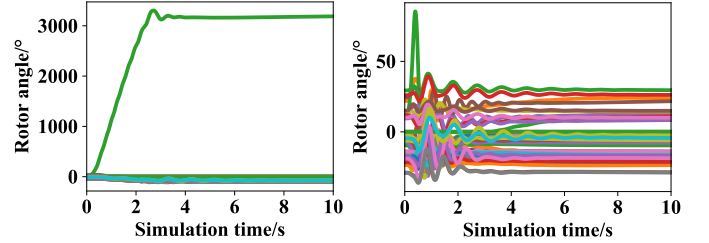


Fig. 9. Rotor angle trajectories before and after preventive control without robustness verification.

be observed that, after preventive control, the control strategy  $\mu$  changes the power angle trajectory from divergent to convergent, transitioning from transient instability to stability. However, when considering perturbations of  $\tilde{x}$ , there may be unstable points within  $\mathcal{C}$ . To verify this, set the maximum perturbation range for IBRs to 20%, for SGs to 8%, and for loads to 10%, to construct  $\mathcal{C}$ . It is important to note that in Fig. 6, when the perturbation reaches 10%, almost all samples are classified as unsafe-PGD. However, a larger perturbation, such as 20% for IBRs, can be set here because the safety and robustness of the preventive control strategies from TSC-OPF have been significantly improved overall. Then, MCS is used to randomly sample points from  $\mathcal{C}$  and verify their power angle trajectories. After hundreds of MCS samples, two adversarial samples  $x_{adv}$  are found, as shown in Fig. 10, indicating the potential risks of directly applying the initial preventive control strategy.

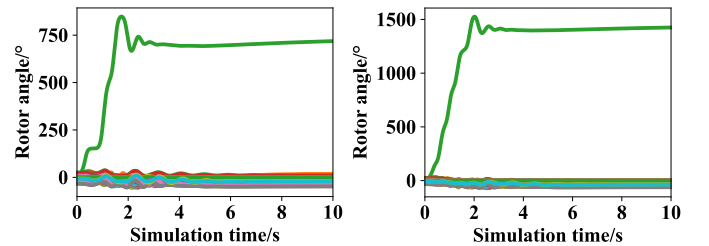


Fig. 10. Rotor angle trajectories of  $x_{adv}$  without robustness verification.

Next, using the same  $\mathcal{C}$ , TSC-OPF is performed according to Algorithm 2, and the bisection method is used to find the optimal  $\lambda$  when robustness verification fails. The iterative process is shown in Table III. From Table III, it can be observed that when  $\lambda$  is too large, there is a risk of TSC-OPF being diverged. This occurs because the safety margin for transient stability constraints becomes excessively large, resulting in no feasible solution. When  $\lambda \leq 61.875$ , as in iterations 1, 2, 4, and 5, the safety margin is insufficient, and the PGD attack succeeds, indicating that these  $\tilde{x}$  are unsafe. When  $\lambda = 67.5$ , as in iteration 3, even though

TABLE III  
ITERATIVE PROCESS OF BISECTION METHOD TO FIND THE OPTIMAL  $\lambda$

| It. | $\lambda$ | Converge | Verification  | Cost/\$ | TSI      |
|-----|-----------|----------|---------------|---------|----------|
| 1   | 0.000000  | True     | unsafe-PGD    | 19035.6 | 61.26032 |
| 2   | 45.000000 | True     | unsafe-PGD    | 19035.6 | 61.26032 |
| 3   | 67.500000 | False    | safe-complete | 20431.9 | 65.75372 |
| 4   | 56.250000 | True     | unsafe-PGD    | 19035.6 | 61.26032 |
| 5   | 61.875000 | True     | unsafe-PGD    | 19037.5 | 61.87499 |
| 6   | 64.687500 | True     | safe-complete | 19102.4 | 64.68749 |
| 7   | 63.28125  | True     | safe-complete | 19057.8 | 63.28125 |

TSC-OPF does not converge, the verification result is safe-complete. This occurs because, while the transient stability constraint is satisfied, other constraints within TSC-OPF are not, leading to non-convergence. From the cost and TSI, it can be seen that in some iterations, even though  $\lambda$  is different, the cost and TSI remain unchanged, such as for  $\lambda = 0, 45$ , and  $56.25$  in iterations 1, 2, and 4. This is because  $\lambda$  is relatively small ( $\lambda < 61.26032$ ), causing the TSC-OPF to be limited by other constraints rather than the TSI constraint. Finally, after 7 iterations, the bisection method converges with  $\lambda = \text{TSI} = 63.28125$ , indicating that the TSC-OPF has fully converged to the boundary of the transient stability constraint, with robustness also verified as safe. Compared to not performing NN robustness verification, the cost increased by only \$22.2 (from \$19,035.6 to \$19,057.8) after executing NN robustness verification. At this time, the active power generation of IBRs and SGs, compared to  $x$  without preventive control, is shown in Fig. 11. By comparing Figs. 8 and 11, we

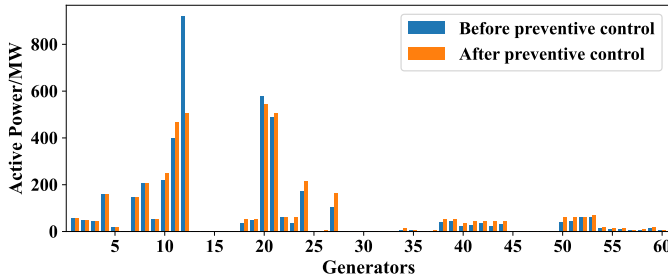


Fig. 11. Comparison of active power generation of IBRs and SGs before and after preventive control with NN robustness verification.

find that the generation of most generators remains identical, except for generators 12 and 25, where 25.53 MW is shifted from generator 12 to generator 25 to satisfy the requirement of robustness verification. The rotor angle trajectories before and after preventive control are shown in Fig. 12. Undoubtedly, the verified  $\tilde{x}$  can ensure that the system maintains transient stability. Finally, to further verify the robustness of points within  $\mathcal{C}$ , similar to Fig. 10, MCS is used to generate a large number of  $x$  within  $\mathcal{C}$  and verify their power angle trajectories. Since all trajectories are stable, only two power angle trajectories are shown in Fig. 13, demonstrating that  $\alpha, \beta$ -CROWN is a complete robustness verification method.

## V. CONCLUSION

This paper proposes an NN robustness verification-based power system transient stability preventive control method

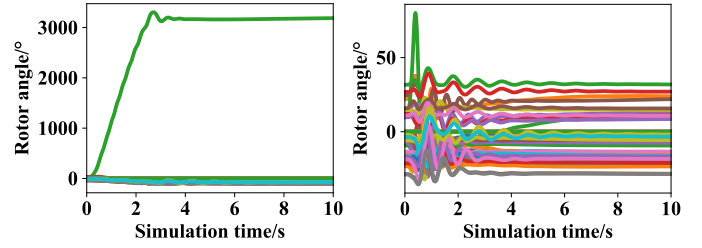


Fig. 12. Rotor angle trajectories before and after preventive control with robustness verification.

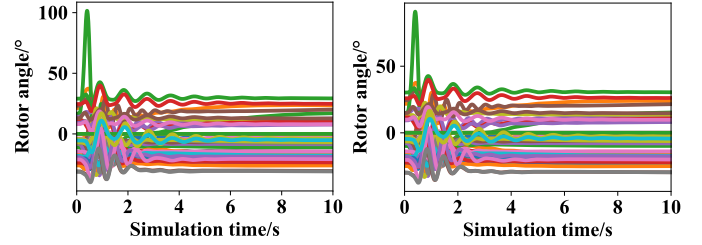


Fig. 13. Rotor angle trajectories of points in  $\mathcal{C}$  with robustness verification.

considering errors from various measurement devices, fluctuations in RESs and loads, as well as cyber-attacks.  $\alpha, \beta$ -CROWN-informed NN certification is developed to verify the robustness of the preventive control strategy within a maximum possible perturbation range. If the verification is unsafe, the bisection method is used to increase the transient stability boundary, thereby increasing the safety margin. The proposed method can balance system security and economics while being scalable to large-scale systems. Numerical results on a modified western South Carolina 500-bus system demonstrate that the proposed method can efficiently and quickly get the safety-verified preventive control strategy through RES curtailment and SG dispatch with only a slight increase in cost. In future research, we will consider incorporating the defense against a broader range of cyber-attacks into the proposed framework.

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